



Effect of seed treatment duration on growth and colonization of *Vicia faba* by endophytic *Beauveria bassiana* and *Metarhizium brunneum*

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ABSTRACT

The effect of increased duration of seed treatment with two fungal entomopathogens, *Beauveria bassiana* (NATURALIS) and *Metarhizium brunneum* (BIPESCO5), on the growth and colonization of broad bean (*Vicia faba*) was assessed in two replicate greenhouse experiments. The plants were inoculated by soaking seeds in a concentration of 1×10^7 conidia ml^{-1} of each of the two fungal strains or in sterile distilled water containing 0.1% Tween 80 for 2, 8, and 16 h. Percent seed germination and seedling emergence as well as primary root length were determined seven days post inoculation. Effects of fungal strain and seed treatment duration on plant height, number of leaf pair, fresh root and shoot weight were measured 14 and 28 days post inoculation. Seed treatment with either fungal strain significantly enhanced almost all measured plant growth parameters. Overall, plant growth was also significantly enhanced as seed treatment duration increased; irrespective of fungal strain. Percent colonization of different plant parts (root, leaf, and stem) sampled 14 and 28 days post inoculation increased as the duration of seed treatment increased as well. The increase in colonization rate due to increased seed treatment duration varied significantly among plant parts, but not among both sampling dates. Our study demonstrates that *M. brunneum* can systemically colonize different plant parts and improve plant growth when applied as a seed treatment. This was also confirmed for the well-known endophytic *B. bassiana* strain used in this study. Our results also provide the first report for the effect of increased duration of seed treatment with both endophytic fungal entomopathogens on plant growth enhancement.

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1. Introduction

Several species of fungal entomopathogens are now known to have complex relationships with plants as endophytes (Vega, 2008; Gurulingappa et al., 2010; Parsa et al., 2013), plant disease antagonists (Goettel et al., 2008; Ownley et al., 2010), rhizosphere colonizers (Hu and St. Leger, 2002; St. Leger, 2008), and plant growth promoters (Liao et al., 2014; Lopez and Sword, 2015). That entomopathogenic fungi have a range of ecological functions suggests that these fungi may have the potential to be developed as biopesticides with multiple roles for improved production of agricultural systems throughout the world (Vega et al., 2009; Lacey et al., 2015). Yet, most studies exploring the multifunctional potential of fungal entomopathogens have focused so far on the endophytic ability of these microbial agents to colonize plants and act against insect and pathogen pests (e.g. Akello et al., 2008; Ownley et al., 2008; Quesada-Moraga et al., 2009; Gurulingappa et al., 2010, 2011; Akello and Sikora, 2012; Akutse et al., 2013; Castillo-Lopez et al., 2014; Jaber and Salem, 2014; Jaber, 2015; Mantzoukas et al., 2015; Vidal and Jaber, 2015). To date, little attention has been given to their possible role as plant growth promoting agents and very few studies have demonstrated the ability of fungal entomopathogens to improve plant growth following their endophytic colonization of

plants (Garcia et al., 2011; Sasan and Bidochka, 2012; Lopez and Sword, 2015).

Fungal endophytes are commonly defined as fungi that colonize the internal tissues of plants for some or all of their lifecycle without causing visible signs of infection (Wilson, 1995). Plant colonization by fungal endophytes can either be localized or systemic (Yan et al., 2015; and references therein). Among fungal entomopathogens, *Beauveria bassiana* (Balsamo) Vuillemin (Ascomycota: Hypocreales) is the most widely researched as an endophyte (Parsa et al., 2013). It has been reported to colonize a variety of host plants, both naturally and artificially following the use of various inoculation methods (e.g. Akello et al., 2007; Posada et al., 2007; Vega, 2008; Tefera and Vidal, 2009; Brownbridge et al., 2012; Russo et al., 2015). Furthermore, Lopez and Sword (2015) recently demonstrated that endophytic *B. bassiana* enhanced cotton growth when plants were inoculated with the fungus as seed treatment and grown under greenhouse conditions. On the other hand, fungal entomopathogens in the genus *Metarhizium* are less characterized as endophytes; although the principle habitat of *Metarhizium* spp. is not insects, but the rhizosphere where different species and strains of *Metarhizium* form various associations with different plant species (St. Leger et al., 2011). More recently however, there appears to be a growing appreciation of the relationship between *Metarhizium* and the endosphere of plants with an increasing number of studies reporting the presence of several species of *Metarhizium*, such as *Metarhizium anisopliae* and *Metarhizium robertsii*, as endophytes in different plant species (Garcia et al., 2011; Akello and Sikora, 2012; Sasan and Bidochka, 2012; Behie et al.,

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2015; Greenfield et al., 2016). To the best of our knowledge, the endophytic capacity of *Metarhizium brunneum* Petch (Ascomycota: Hypocreales) has not been previously demonstrated following its artificial inoculation into plants through seeds (however, see Resquin-Romero et al., 2016). Similarly, effects of seed treatment duration with endophytic fungal entomopathogens on the systemic colonization and growth promotion of plants has not been previously investigated. Therefore, we conducted greenhouse experiments to determine (1) if *M. brunneum* can be established as a systemic endophyte in broad bean (*Vicia faba*) when applied as a seed treatment compared to an endophytic strain of *B. bassiana* already known to systemically colonize broad bean and other plants, (2) if inoculation with these two fungal entomopathogens affects plant growth, and (3) if increased duration of seed treatment results in increased fungal colonization and plant growth.

2. Materials and methods

2.1. Plant and fungal material

Broad bean (*V. faba*) was selected as the model host plant for our study, because previous studies have shown that it can successfully be colonized by endophytic fungal entomopathogens (Akello and Sikora, 2012; Akutse et al., 2013; Vidal and Jaber, 2015). Seeds of broad bean (cv. Cobrocy) were used for all experiments. Prior to use, seeds were surface-sterilized by soaking in 1% sodium hypochlorite (NaOCl) for two minutes followed by soaking in 70% ethanol for two minutes, and finally rinsing in sterile distilled water three times. Samples taken from the third rinse water were plated on potato dextrose agar (PDA) media to confirm the efficiency of the surface sterilization procedure (Schulz et al., 1998). *Metarhizium brunneum* strain BIPESCO5, which is registered for pest biocontrol in Europe, was selected because it has never been previously reported as an endophyte. *Beauveria bassiana* strain ATCC 74040, registered as NATURALIS® (Intrachem Bio Italia S.p.A., Bergamo, Italy), was also used in our study. This strain has been shown to colonize different host plants as an endophyte (Jaber and Salem, 2014; Jaber, 2015; Vidal and Jaber, 2015). Both fungal strains were cultured and maintained on PDA plates at 25 °C in complete darkness.

2.2. Plant inoculation with fungal entomopathogens

Fungal conidia were harvested under sterile conditions by gently scraping the surface of 14-day-old sporulating cultures of *B. bassiana* and *M. brunneum* after flooding the plates with sterile water containing Tween 80 (0.1% v/v, Difco™). The resulting mycelia and spores were filtered through several layers of sterile cheese cloth into a sterile glass bottle containing 400 ml sterile distilled water plus 0.1% Tween 80. The conidial suspension of each fungal strain was then homogenized with a magnetic stirrer for 5 min. The concentration of conidia in each fungal suspension was determined under the microscope using a Neubauer hemocytometer and adjusted to a concentration of 1×10^7 conidia ml⁻¹. This concentration has been previously used to establish different fungal entomopathogens as endophytes (Posada and Vega, 2005; Akello et al., 2007; Gurulingappa et al., 2010, 2011; Castillo-Lopez et al., 2014; Lopez and Sword, 2015). Conidial viability for each fungal strain was determined prior to inoculation trials by plating 100 µl aliquot of a dilute conidial suspension onto PDA and incubating at 25 °C for 24 h. Germination was then checked under the microscope and found to exceed 90% for each strain on average. Surface-sterilized seeds were then soaked in the conidial suspension of each fungal strain or in sterile distilled water

containing 0.1% Tween 80 (control treatment) for 2, 8, and 16 h. Bottles containing the soaking seeds were kept in the dark at 25 °C for the designated duration. Inoculated and control seeds were then either used in a germination test or transferred into disinfected plastic pots containing autoclaved planting substrate for further bioassays with growth parameters (as described below in Section 2.3).

2.3. Effect of fungal inoculation by increased seed treatment duration on plant growth

To determine if increased seed treatment duration has an influence on seed germination, five seeds soaked in the conidial suspension of each fungal strain or in sterile distilled water containing 0.1% Tween 80 (control treatment) were placed on 9 cm Petri dishes lined with sterile moist filter paper. Sterile water was added as needed to keep the filter paper moist. Ten replicate dishes were prepared for each treatment combination (see below) and the Petri dishes were kept in the dark at 20 °C for a week. A total of 50 seeds were used per treatment combination. The total germination percentage and the length of primary root were recorded seven days post inoculation.

To examine effects of increased seed treatment duration on additional plant growth parameters, inoculated or control seeds were planted in disinfected pots containing sterile planting substrate (mixture of soil, sand, and peat at a ratio of 1:1:1). The planting substrate was sterilized by autoclaving at 121 °C for 2 h and allowed to cool for 24 h prior to potting. Two seeds were sowed per pot. Potted plants were maintained in the greenhouse at 25 ± 3 °C, 60% relative humidity, and a photoperiod of 16:8 h light/dark. Plants were watered with sterile distilled water as needed, and were not fertilized throughout the experiments. Percent seedling emergence was measured seven days post inoculation. After emergence, seedlings were thinned to one per pot and allowed to grow for 28 days. Plant height (the distance from plant base to the tip of the stem), number of unfolded leaf pairs, fresh shoot weight, and fresh root weight were measured 14 and 28 days post inoculation. The experiments were organized as a complete randomized design with a factorial arrangement. Factors included fungal strain inoculation (*M. brunneum* BIPESCO5, *B. bassiana* NATURALIS, and the control) and seed treatment duration (2, 8, and 16 h). A total of nine treatment combinations (fungal strain inoculation × seed treatment duration) were used. Each treatment combination had a total of 20 plants, ten of which were destructively sampled for plant growth parameters and fungal colonization (see Section 2.4) 14 days post inoculation and the remaining ten were sampled 28 days post inoculation. All experiments were independently replicated twice.

2.4. Effect of fungal inoculation by increased seed treatment duration on the endophytic colonization of plants

Fourteen and 28 days post seed inoculation with *M. brunneum* and *B. bassiana*, local or systemic plant colonization by these two fungal entomopathogens was determined through re-isolation of fungi from different plant parts using the method described by Arnold et al. (2001). Leaf pairs, stems, and roots (after thorough washing with running tap water) of ten plants per treatment combination were processed on each sampling date. The same plants used for sampling growth parameters were later used for detecting endophytic colonization by the respective fungi. Plant material was surface-sterilized by submerging in 0.5% sodium hypochlorite (for leaf and stem tissues) and in 1% sodium hypochlorite (for root tissues) for two minutes, followed by two minutes in 70% ethanol and three rinses in sterile distilled water. Plant material was then allowed to surface-dry in a laminar flow hood. An average of eight leaf discs (approximately 2 mm²)

were cut from each surface-sterilized leaf pair using a sterile cork borer and a similar number of stem tissue pieces (approximately 8–10 mm long) were obtained by a sterile scalpel. Roots were first divided into proximal and distal root ends. Proximal and distal ends of the roots were then divided into primary roots, lateral roots, and root hairs. An average of eight randomly selected segments (approximately 8–10 mm long) of the different root parts were cut from each plant replicate using a sterile scalpel. Sampled plant tissues were then plated on 9 cm Petri dishes containing semi-selective culture medium (Strasser et al., 1996) that was modified according to Posadas et al. (2012). Plant tissue imprints were obtained prior to plating and the last rinse water was also plated out to assess the effectiveness of the surface sterilization procedure (Schulz et al., 1998). Plates were incubated at 25 °C and inspected every 2–3 days for three weeks to observe and record fungal outgrowth. Fungal outgrowth from the plated plant samples were identified as *M. brunneum* or *B. bassiana* based on differential growth on the semi-selective media, colony morphology, and microscopic examination of conidia (Humber, 1997). Percent colonization of different plant parts by the respective inoculated fungus was calculated following the Petrini and Fisher (1987) formula: % colonization = number of sampled plant tissue showing fungal outgrowth divided by the total number of plated plant tissue samples $\times$ 100. Presence of other bacterial or fungal endophytes on the incubated plates was occasionally observed, but not scored in our study.

2.5. Statistical analyses

Prior to statistical analyses, all data were tested for normality and homogeneity of the variance. Percentage values of seed germination, seedling emergence, and colonization of different plant parts with *M. brunneum* and *B. bassiana* were positively skewed (Skewness > 0) and showed non-Gaussian distributions (Kolmogorov-Smirnov, $P < 0.05$). Therefore, data were transformed using a natural logarithm (ln) function [$\ln(y) = \ln(y + 1)$] to correct for heterogeneity of the variance and produce approximately normally distributed data sets. Data from both experimental replicates were subjected to two-way ANOVA (GLM procedure) with fungal strain inoculation and seed treatment duration as the main factors. When a significant F test was obtained at $P = 0.05$, separation of treatment means was performed using Duncan's Multiple Range Test. The Statistical Analysis System (SAS) software Version 9.1.3 (SAS Institute Inc., 2010) was used for conducting all statistical analyses.

3. Results

3.1. Effect of fungal inoculation by increased seed treatment duration on plant growth

Both main factors, fungal strain inoculation and seed treatment duration, had significant effects on almost all of the plant growth parameters sampled 7, 14 and 28 days post inoculation (dpi) in both experimental replicates (Figs. 1–4). Inoculation with either fungal entomopathogen, *M. brunneum* or *B. bassiana*, significantly enhanced percent seedling emergence ($F = 5.10$; $df = 2$; $P = 0.0071$ in experiment 2) (Fig. 1B), plant height ($F = 10.56$; $df = 2$; $P < 0.0001$ at 14 dpi, $F = 12.76$; $df = 2$; $P < 0.0001$ at 28 dpi, $F = 11.22$; $df = 2$; $P < 0.0001$ at 14 dpi, $F = 4.82$; $df = 2$; $P = 0.0109$ at 28 dpi in experiments 1 and 2, respectively), number of leaf pairs ($F = 14.60$; $df = 2$; $P < 0.0001$ at 28 dpi in experiment 1, $F = 10.72$; $df = 2$; $P < 0.0001$ at 14 dpi, $F = 13.67$; $df = 2$; $P < 0.0001$ at 28 dpi in experiment 2), fresh shoot weight ($F = 14.36$; $df = 2$; $P < 0.0001$ at 14 dpi, $F = 4.68$;

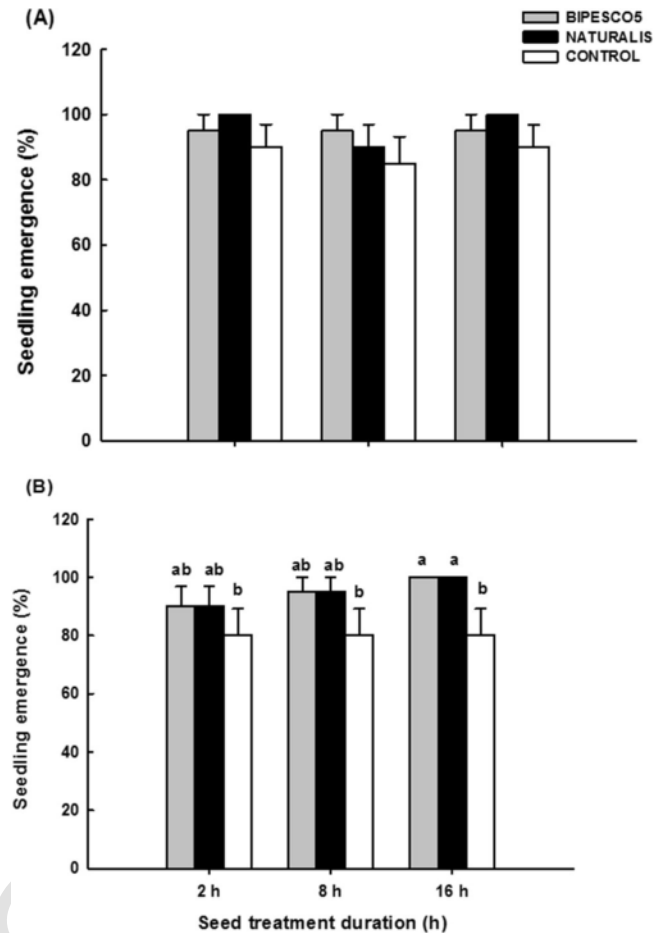


Fig. 1. Effect of fungal strain inoculation (*Metarhizium brunneum* strain BIPESCO5, *Beauveria bassiana* strain NATURALIS, and the control) and seed treatment duration (2, 8, and 16 h) on mean ($\pm$ SE) percent seedling emergence of broad bean (*Vicia faba*) seven days post inoculation (dpi). Bars with different letters differ significantly among all seed treatment durations at $P = 0.05$ (Duncan's Multiple Range Test after two-way ANOVA). Letters A and B denote experimental replicates 1 and 2.

$df = 2$; $P = 0.0120$ at 28 dpi in experiment 1, $F = 15.98$; $df = 2$; $P < 0.0001$ at 14 dpi in experiment 2), and fresh root weight ($F = 3.61$; $df = 2$; $P = 0.0319$ at 14 dpi in experiment 1 and $F = 5.33$; $df = 2$; $P = 0.0069$ at 14 dpi in experiment 2) (Figs. 2 and 3). Irrespective of the fungal strain being inoculated, increasing seed treatment duration had a significant effect on plant height measured at 14 dpi in experiment 2 ($F = 4.85$; $df = 2$; $P = 0.0105$), number of leaf pairs at 28 dpi in experiment 2 ($F = 3.43$; $df = 2$; $P = 0.0379$), and fresh shoot weight at 14 dpi ($F = 7.07$; $df = 2$; $P = 0.0015$, $F = 4.09$; $df = 2$; $P = 0.0208$) in experiments 1 and 2, respectively (Figs. 2 and 3). Increasing seed treatment duration resulted in a marginally significant increase in the fresh root weight of inoculated plants only at 14 dpi in experiment 2 ($F = 2.47$; $df = 2$; $P = 0.0914$, Fig. 3).

For the seed germination test, both fungal strain and seed treatment duration had significant effects on primary root length ($F = 3.57$; $df = 2$; $P = 0.0289$, $F = 7.56$; $df = 2$; $P = 0.0006$ for fungal strain in experiments 1 and 2, respectively, $F = 10.08$; $df = 2$; $P < 0.0001$, $F = 6.14$; $df = 2$; $P = 0.0023$ for seed treatment duration in experiments 1 and 2, respectively) (Fig. 4), but percent seed germination was not affected (data not shown). Generally, all measured growth parameters were significantly enhanced in plants grown from seeds soaked in the suspension of either fungal strain for 16 h com-

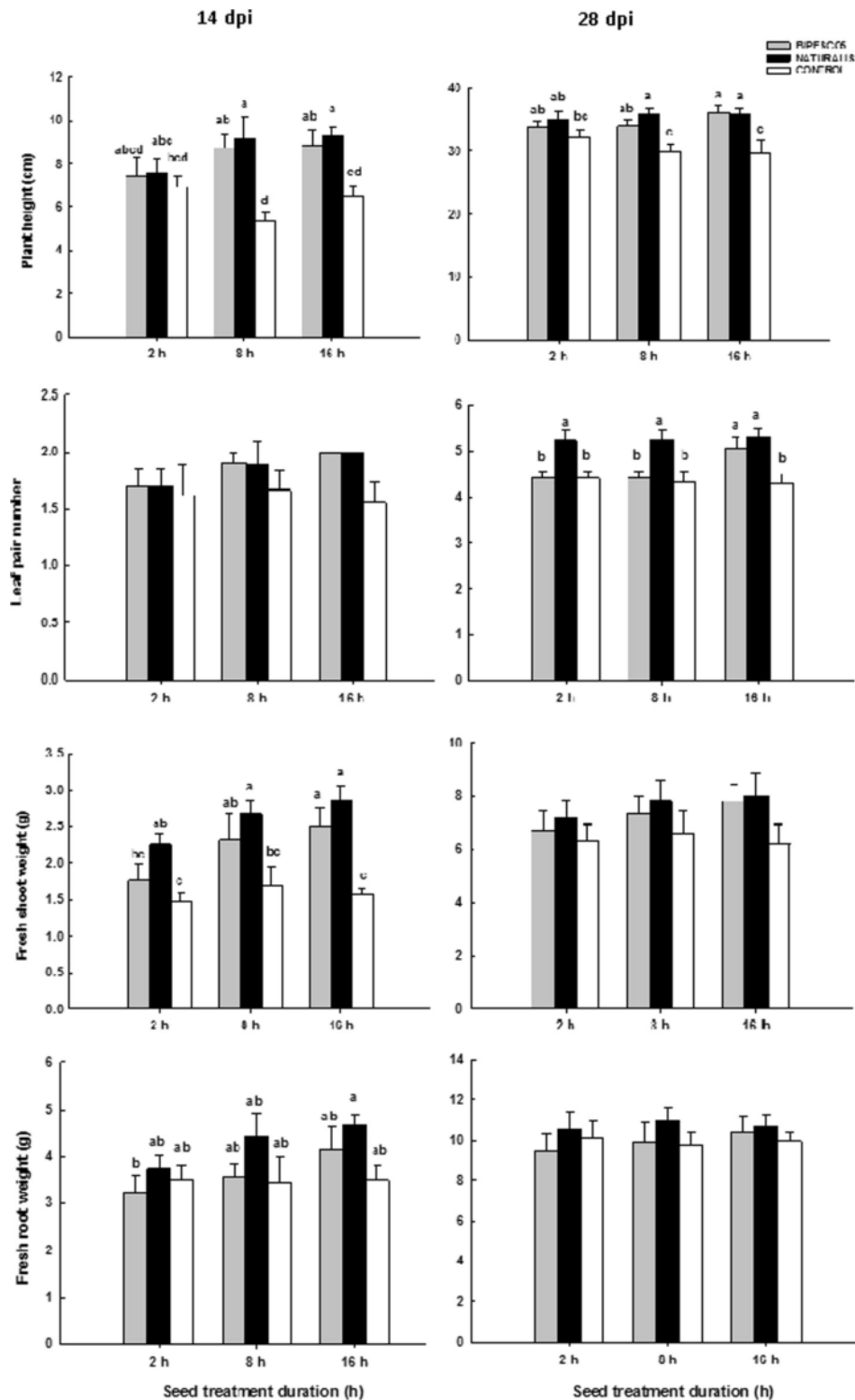


Fig. 2. Effect of fungal strain inoculation (*Metarhizium brunneum* strain BIPESCO5, *Beauveria bassiana* strain NATURALIS, and the control) and seed treatment duration (2, 8, and 16 h) on mean ($\pm$ SE) plant height, number of leaf pair, and fresh shoot and root weight of broad bean (*Vicia faba*) sampled 14 and 28 days post inoculation (dpi) in the first experimental replicate. Bars with different letters differ significantly among all seed treatment durations at $P = 0.05$ (Duncan's Multiple Range Test after two-way ANOVA).

pared to plants grown from seeds soaked for 2 h ($P = 0.05$; Duncan's Multiple Range Test after two-way ANOVA). On the other hand,

there were almost no significant differences in growth enhancement

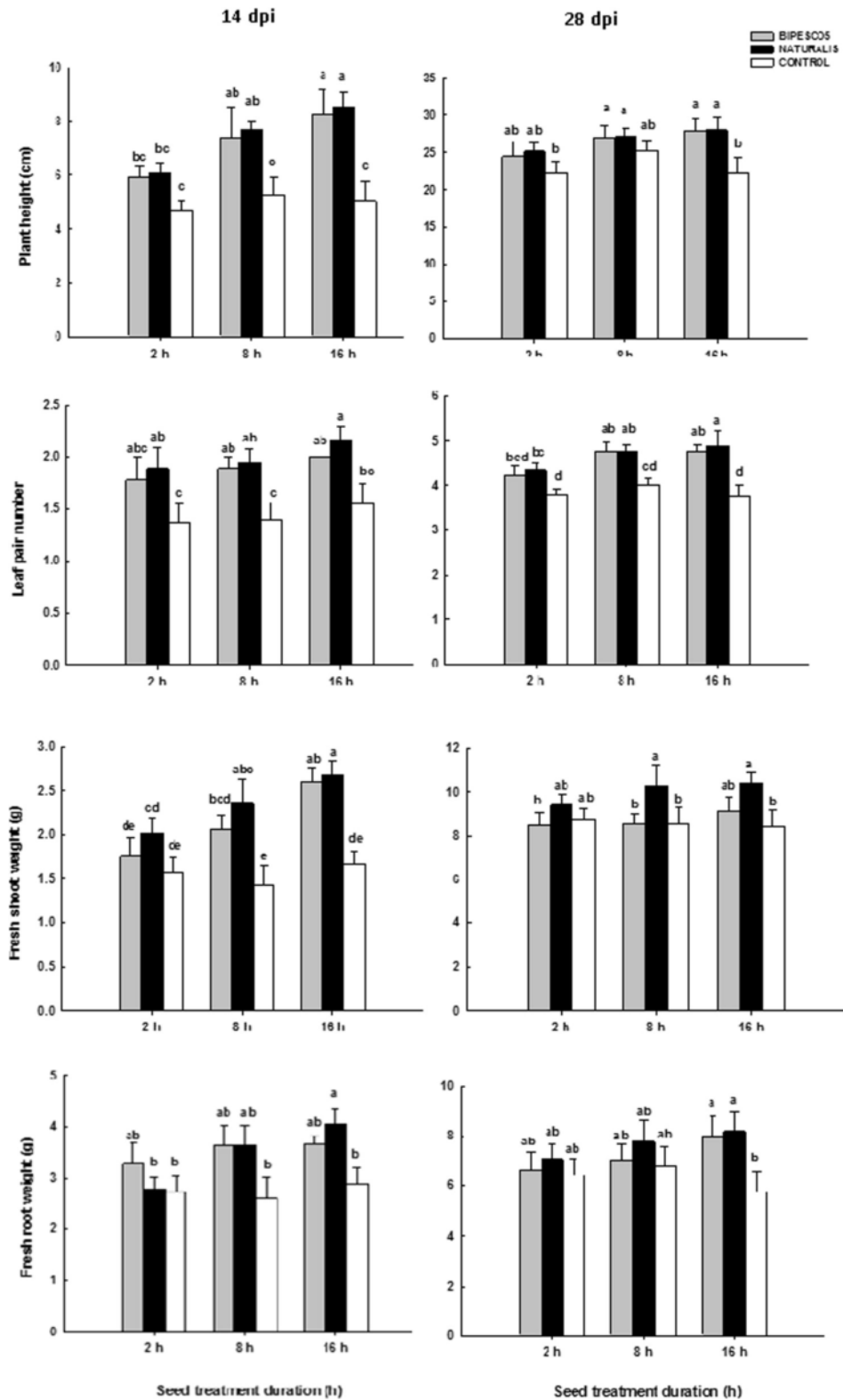


Fig. 3. Effect of fungal strain inoculation (*Metarhizium brunneum* strain BIPESCO5, *Beauveria bassiana* strain NATURALIS, and the control) and seed treatment duration (2, 8, and 16 h) on mean ($\pm$ SE) plant height, number of leaf pair, and fresh shoot and root weight of broad bean (*Vicia faba*) sampled 14 and 28 days post inoculation (dpi) in the second experimental replicate. Bars with different letters differ significantly among all seed treatment durations at $P = 0.05$ (Duncan's Multiple Range Test after two-way ANOVA).

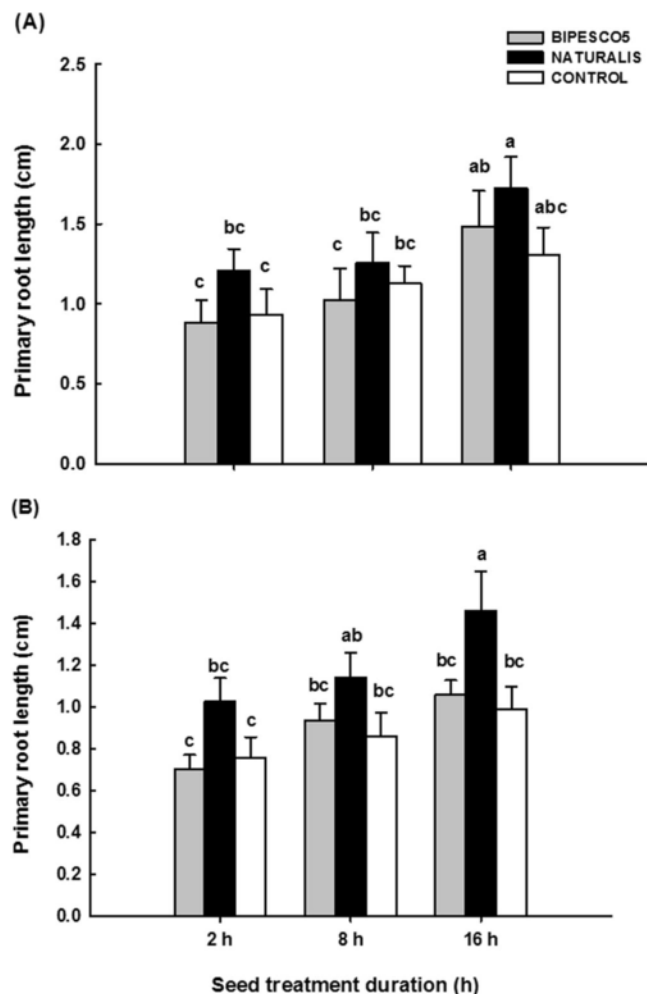


Fig. 4. Effect of fungal strain inoculation (*Metarhizium brunneum* strain BIPESCO5, *Beauveria bassiana* strain NATURALIS, and the control) and seed treatment duration (2, 8, and 16 h) on mean ($\pm$ SE) primary root length of broad bean (*Vicia faba*) seven days post inoculation (dpi). Bars with different letters differ significantly among all seed treatment durations at $P = 0.05$ (Duncan's Multiple Range Test after two-way ANOVA). Letters A and B denote experimental replicates 1 and 2.

of plants grown from seeds that were soaked for 8 or 16 h (Figs. 1–4).

Table 1

Effect of fungal strain inoculation (*Beauveria bassiana* strain NATURALIS, *Metarhizium brunneum* strain BIPESCO5, and the control) and seed treatment duration (2, 8, and 16 h) on mean ($\pm$ SE) percentage colonization of different plant parts (root, stem, and leaf) of broad bean (*Vicia faba*) 14 and 28 days post inoculation (dpi). Experimental replicates 1 and 2 are presented.

Experiment	Plant part	<i>Beauveria bassiana</i> (NATURALIS)						<i>Metarhizium brunneum</i> (BIPESCO5)					
		2 h		8 h		16 h		2 h		8 h		16 h	
		14 dpi	28 dpi	14 dpi	28 dpi	14 dpi	28 dpi	14 dpi	28 dpi	14 dpi	28 dpi	14 dpi	28 dpi
1	Root	22 $\pm$ 6bc	17 $\pm$ 9bc	33 $\pm$ 14ab	31 $\pm$ 13ab	51 $\pm$ 6a	46 $\pm$ 12ab	25 $\pm$ 7b	27 $\pm$ 13abc	37 $\pm$ 11ab	54 $\pm$ 15a	56 $\pm$ 8a	58 $\pm$ 13;
	Stem	32 $\pm$ 16bc	27 $\pm$ 13 cd	44 $\pm$ 9ab	39 $\pm$ 17bc	66 $\pm$ 8a	64 $\pm$ 10a	30 $\pm$ 13bc	10 $\pm$ 3 cd	35 $\pm$ 15ab	31 $\pm$ 14bc	54 $\pm$ 16ab	51 $\pm$ 12;
	Leaf	23 $\pm$ 10ab	21 $\pm$ 10abc	28 $\pm$ 10ab	27 $\pm$ 12ab	32 $\pm$ 14a	33 $\pm$ 12a	15 $\pm$ 7ab	8 $\pm$ 3bc	18 $\pm$ 10ab	18 $\pm$ 10abc	23 $\pm$ 10ab	20 $\pm$ 7a
2	Root	24 $\pm$ 6ab	20 $\pm$ 10bc	40 $\pm$ 17a	37 $\pm$ 15ab	49 $\pm$ 13a	39 $\pm$ 12ab	25 $\pm$ 11ab	28 $\pm$ 15abc	39 $\pm$ 14a	40 $\pm$ 11ab	53 $\pm$ 15a	56 $\pm$ 16;
	Stem	29 $\pm$ 15ab	25 $\pm$ 8ab	41 $\pm$ 14a	37 $\pm$ 9a	49 $\pm$ 13a	45 $\pm$ 18a	27 $\pm$ 13ab	25 $\pm$ 15ab	39 $\pm$ 14a	38 $\pm$ 12a	48 $\pm$ 19a	41 $\pm$ 12;
	Leaf	23 $\pm$ 10ab	20 $\pm$ 11ab	30 $\pm$ 14ab	27 $\pm$ 14a	34 $\pm$ 16a	32 $\pm$ 10a	11 $\pm$ 5ab	0 $\pm$ 0b	20 $\pm$ 8ab	13 $\pm$ 6ab	25 $\pm$ 12ab	24 $\pm$ 11;

For each plant part, mean ($\pm$ SE) followed by the same letter within the same sampling date are not significantly different at $P = 0.05$ (Duncan's Multiple Range Test after two-way ANOVA). Among both sampling dates, mean ($\pm$ SE) percentage colonization of different plant parts with *B. bassiana* or *M. brunneum* was not significantly different across treatments ($P = 0.05$; Duncan's Multiple Range Test after two-way ANOVA). Since neither *B. bassiana* nor *M. brunneum* was ever isolated from control plants, data obtained from those plants are not shown.

Moreover, there were no significant interactions between fungal strain and seed treatment duration for any of the measured plant growth parameters.

3.2. Effect of fungal inoculation by increased seed treatment duration on the endophytic colonization of plants

Plant colonization by *M. brunneum* and *B. bassiana* was determined 14 and 28 days post seed treatment through re-isolating fungi on culture media. All inoculated plants were colonized by the respective fungi on both sampling dates. Neither *M. brunneum* nor *B. bassiana* were isolated from the control plants. No fungal colonies were detected following inoculation of plates with the final rinse water or on imprint plates, indicating the efficacy of the surface-sterilization procedure and confirming that the fungi growing from the surface-sterilized plant materials originated from within plant tissues, i.e. were endophytic. Other fungi and bacteria were occasionally observed growing from sampled tissues of inoculated and control plants. However, only the presence of the two fungal entomopathogens that were inoculated into plants through seed treatment was scored in our study.

Table 1 shows the effect of fungal strain inoculation and seed treatment duration on percent colonization of different plant parts (root, stem, and leaf) by *M. brunneum* and *B. bassiana* 14 and 28 days post inoculation (dpi). Increasing seed treatment duration increased the percent colonization of plant root, stem, and leaf tissues with both fungal strains on both sampling dates and in each experimental replicate. However, the increase in percent plant colonization in response to increased seed treatment duration was only significant for roots at 14 dpi ($F = 5.41$; $df = 2$; $P = 0.0088$) and stems at 28 dpi ($F = 4.98$; $df = 2$; $P = 0.0123$) in the first experimental replicate (Table 1). The increased percent colonization of roots at 28 dpi and stems at 14 dpi of the first experimental replicate was only marginally significant ($F = 2.88$; $df = 2$; $P = 0.0689$ and $F = 2.59$; $df = 2$; $P = 0.0892$, respectively). There was no significant interaction between fungal strain and seed treatment duration on colonization of any of the sampled plant parts. On the other hand, the increase in percent colonization due to increased seed treatment duration varied significantly among plant parts within each sampling date ($F = 5.55$; $df = 2$; $P = 0.0049$ at 14 dpi, $F = 4.31$; $df = 2$; $P = 0.0156$ at 28 dpi, $F = 2.80$; $df = 2$; $P = 0.0646$ at 14 dpi, $F = 4.04$; $df = 2$; $P = 0.0199$ at 28 dpi for experiments 1 and 2, respectively). Also, the plant tissue colonized varied according to the fungal strain used to inoculate the

seeds, implying that the fungal strains colonized plant parts differently (Table 1). For example, *M. brunneum* colonized more roots (56% at 14 dpi, 58% at 28 dpi and 53% at 14 dpi, 56% at 28 dpi in experiments 1 and 2, respectively) than stems (54% at 14 dpi, 51% at 28 dpi and 48% at 14 dpi, 41% at 28 dpi in experiments 1 and 2, respectively) and leaves (23% at 14 dpi, 20% at 28 dpi and 25% at 14 dpi, 24% at 28 dpi in experiments 1 and 2, respectively) of plants grown from seeds inoculated for 16 h; whereas *B. bassiana* colonized more stems (66% at 14 dpi, 64% at 28 dpi and 49% at 14 dpi, 45% at 28 dpi in experiments 1 and 2, respectively) than roots (51% at 14 dpi, 46% at 28 dpi and 49% at 14 dpi, 39% at 28 dpi in experiments 1 and 2, respectively) or leaves (32% at 14 dpi, 33% at 28 dpi and 34% at 14 dpi, 32% at 28 dpi in experiments 1 and 2, respectively) of plants grown from seeds inoculated for 16 h. Percent colonization of different plant parts by *B. bassiana* and *M. brunneum* was not significantly different across treatments between both sampling dates. However irrespective of seed treatment duration, a decline in percent colonization was frequently observed over time for different plant parts inoculated with *B. bassiana*; whereas the decline in percent colonization over time was only observed in stems and leaves but not roots of plants inoculated with *M. brunneum* (Table 1).

4. Discussion

Our study demonstrates the potential of *M. brunneum* to systemically colonize different parts of *V. faba* and improve plant growth after its application as seed treatment. Similar results were obtained for *B. bassiana* whose endophytic capacity has been well-documented. Recently, Resquin-Romero et al. (2016) reported that entomopathogenic fungal endophytes, including *B. bassiana* and *M. brunneum*, only temporarily colonize their host plants in a non-systemic pattern after spray application. The authors argue that such a transient non-systemic plant colonization should be expected in sprayed areas since foliar application of the fungal inoculum does not usually result in the colonization of the entire plant. Similarly, Posada et al. (2007) suggest that application by leaf spray is a poor method of obtaining *B. bassiana* colonization of coffee plants. However, the fungal strains tested by Resquin-Romero et al. (2016) were able to endophytically colonize leaves, stems, and even roots for the first 96 h after spray application. To date, several inoculation methods have been tested as means whereby fungal entomopathogens can be introduced into plants and different success rates have been achieved according to the method of inoculation. These methods include foliar spray, stem injection, soil drench, root dip, and seed treatment (Akello et al., 2007; Posada et al., 2007; Tefera and Vidal, 2009; Brownbridge et al., 2012; Russo et al., 2015). When seed treatment was used as the inoculation method, *B. bassiana* systemically colonized all tissues of different host plants as shown by several previous studies (Tefera and Vidal, 2009; Brownbridge et al., 2012; Akutse et al., 2013; Landa et al., 2013; Quesada-Moraga et al., 2014). Here, we show similar results not only for *B. bassiana*, but also for *M. brunneum*. Both fungi could be retained by different plant parts for at least one month after seed treatment. Previous studies have reported the endophytic colonization of plants with *B. bassiana* for as long as three months in jute (Biswas et al., 2012), eight months in coffee (Posada et al., 2007), and nine months in radiata pine (Brownbridge et al., 2012). It remains to be tested whether the endophytic colonization of *V. faba* by *M. brunneum* and *B. bassiana* persists for more than one month.

The extent and persistence of endophytic colonization may also vary depending on the genotype-specific interaction between plants and endophytes (Saikkonen et al., 2010) and the method used for detection of endophytic plant colonization (Hyde and Soyong, 2008).

Culture-based detection, used in our study, is a conservative technique that underestimates the presence of endophytic fungi in colonized plant tissues as compared to *in planta* molecular-based detection (Posada et al., 2007; Biswas et al., 2012; Jaber, 2015). The inherent flaws of the surface-sterilization procedure of sampled plant tissues and the competition with faster growing endophytes on culture media might also affect the recovery success of fungal isolates from within plant tissues. In spite of the low sensitivity of the culture-based technique used for detecting plant colonization with the respective inoculated endophytes in the present study, our results demonstrate an increase in percent colonization of plant root, stem, and leaf tissues with both fungal strains in response to increased seed treatment duration. These findings reflect those of an earlier study (Akello et al., 2009), which found that percent colonization of tissue cultured banana was affected by the duration that plant roots were immersed in a suspension of *B. bassiana*. The study found that prolonged exposure (4 h) to fungal inoculum improved the rate of colonization by endophytic *B. bassiana* when plants were treated with the lowest inoculum dose (1.5×10^5 conidia ml⁻¹). Furthermore, the effect of dipping duration on percent plant colonization varied significantly among plant parts and was more pronounced in plant roots than rhizome and pseudostem base tissues (Akello et al., 2009). In our study, the increase in percent plant colonization due to increased seed treatment duration also varied significantly among different plant parts for both fungal strains within each sampling date. In general, *M. brunneum* was more frequently isolated from root than stem or leaf tissues of inoculated plants, while *B. bassiana* was more frequently isolated from stem than root or leaf tissues of inoculated plants. Similar differential colonization of plant parts was previously demonstrated for *B. bassiana* in different host plants (Posada et al., 2007; Tefera and Vidal, 2009; Gurulingappa et al., 2010; Biswas et al., 2012; Akutse et al., 2013). Moreover, Behie et al. (2015) report that *Metarhizium* spp. were almost exclusively found in roots, whereas *B. bassiana* was found throughout the plant. Endophytic fungi are known to exhibit tissue specificity because they are adapted to particular conditions present in a given plant part (Petrini and Fisher, 1987; Liang-Dong et al., 2008). Consequently, the preferential tissue colonization displayed by *B. bassiana* and *M. brunneum* within host plants could be due to differences in the microbial, biochemical, and physiological conditions of different plant tissues. In nature, *Metarhizium* spp. have been found to be rhizosphere-competent and well-adapted to living in the soil (Hu and St. Leger, 2002; St. Leger, 2008). *Metarhizium robertsii* and *M. brunneum* in particular have been shown to interact intimately with the plant in the rhizosphere where these fungi remain active and proliferate (Bruck, 2010). The rhizosphere therefore appears to improve *Metarhizium* persistence for a prolonged period of time, which consequently may have facilitated its extended association with plant roots as an endophyte. This might explain the increase over time, although not significant, in root colonization with *M. brunneum* but not *B. bassiana* among the two sampling dates.

Our results also provide the first report for the positive plant growth enhancing effect in response to increased duration of seed treatment with *B. bassiana* and *M. brunneum*; even though the effect was not always consistent for all the plant growth parameters measured among both experimental replicates. This could be due to unsuspected variation across replicate trials in any number of factors (e.g. temperature, irrigation water, planting substrate, competition with other naturally occurring endophytes, etc.). Besides, fresh plant parameters, known to be less reliable than dry plant parameters, were used in the present study and might have added to the variability in results among sampling dates or experimental replicates. Our findings contrast with those of Akello et al. (2009) who reported that

growth of tissue cultured banana was not affected by endophytic *B. bassiana*, even when plants were inoculated with the highest inoculum rate (1.5×10^8 conidia ml^{-1}) for the longest dipping duration (4 h). On the other hand, the plant growth enhancement observed in our study is in agreement with Gurulingappa et al. (2010) and Lopez and Sword (2015) who found that endophytic *B. bassiana* significantly increased the growth of wheat and cotton plants, respectively. Such contradiction in results is not surprising since the endophytic abilities of particular fungal strains may be strongly associated with host plant species and even plant species varieties (Akutse et al., 2013). Differences in the endophytic abilities of fungal entomopathogens might also be due to innate characteristics of the tested fungal isolate (Posada et al., 2007). In the present study, seed treatment with *M. brunneum* and *B. bassiana* significantly increased percent seedling emergence, primary root length, plant height, leaf pair number, and fresh shoot and root weight; but not percent seed germination. Similar results have been also obtained when the two fungal strains tested here were inoculated into sweet pepper plants (Jaber, unpublished). Consistent with our findings, corn seed germination was not affected by a *Metarhizium* treatment (Kabaluk and Ericsson, 2007; Liao et al., 2014). However, treatment with *Metarhizium* accelerated the vegetative development of winter wheat (Liao et al., 2014) and the root development of switch grass and haricot beans (Sasan and Bidochka, 2012). Also, Garcia et al. (2011) demonstrated that inoculation with *M. anisopliae* promoted the growth of 28-day-old tomato plants, but not when the lowest inoculation rate (8×10^7 conidia ml^{-1}) was used. Similarly in our study, inoculating seeds with the conidial suspension of both fungal strains for 2 h was not enough to produce the plant growth enhancing effect observed when a seed treatment duration of 16 h was used. Establishing the best inoculation method is crucially important to improve plant colonization with endophytic fungal entomopathogens and benefit plant growth. Equally important is a deeper understanding of the mechanism through which these fungi enhance plant growth. To date, we can only speculate as to the reason behind the plant growth promoting effect of endophytic *B. bassiana* and *M. brunneum*; which is likely to be multifactorial as reported for closely related species of *Trichoderma* (Mukherjee et al., 2013). Recently, increased plant health and productivity was attributed to the nitrogen transferred from insect cadavers to plants by endophytic *B. bassiana* and *Metarhizium* spp. (Behie and Bidochka, 2014). Also, several *Metarhizium* spp. were shown to produce plant-growth promoting auxins on roots (Liao et al., 2014). On the other hand, increased duration of seed treatment (i.e. soaking in the fungal suspension) might have triggered seed imbibition, by causing the seed coat to swell and split, and improved conidial uptake by the seed; thereby enhancing the endophytic establishment and growth promotion by the inoculated fungal strains. At the same time, extended seed treatment duration might have caused conidia to swell and begin to germinate, and produce mucilage that could have facilitated conidial attachment to the seed and enhanced seed loading; resulting in the enhanced plant colonization rate and growth responses observed in the current study. Though the underlying mechanism(s) are still unclear and require further elucidation, our work presents evidence that the fungal entomopathogens *M. brunneum* and *B. bassiana*, not only have endophytic capacity, but can also act as plant growth promoters; and that the plant growth promoting effects increased as the duration of seed treatment increased. Unravelling the full potential of fungal entomopathogens could ultimately improve the commercial use of biocontrol agents as multifaceted tools to protect plants from pests and increase their growth and productivity.

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